Introduction to Data Science: Data pre-processing Tutorial 3

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1. Use for this question the data set `DataOutliers.csv`. Import the data set in R/Python and determine how many outliers this data set has. Explain your answer.

2. Use the data set `Inflation.csv` with monthly inflation (in %) in the Netherlands. Consider the following function f_λ with domain the positive real numbers:

$$f_\lambda(x) = \begin{cases} \frac{x^\lambda - 1}{\lambda} & \text{if } \lambda \neq 0, \\ \log(x) & \text{if } \lambda = 0. \end{cases}$$

   Investigate the effect of λ when using the function f_λ to transform the inflation data.

3. Consider 2 observations, A and B in a 2-dimensional space and

$$A = (2, 1) \text{ and } B = (1, 2).$$

   (a) Determine the standardized versions of A and B and denote them by $(x_{1,1}, x_{1,2})$ and $(x_{2,1}, x_{2,2})$. The sample variance s^2 of a data set x_i for $i = 1, 2, \ldots, n$ can be determined as follows: $s^2 = \frac{1}{n}\sum_{i=1}^{n}(x_i - \bar{x})^2$.

   (b) Determine the eigenvalues of the covariance matrix of the standardized data set $(x_{1,1}, x_{1,2})$ and $(x_{2,1}, x_{2,2})$.

   (c) Determine the principal components $\underline{\alpha}_1$ and $\underline{\alpha}_2$ of the data.

   (d) Show that the principal component are orthogonal.

   (e) The smallest eigenvalue is zero. Explain the intuition of this result.

Solution: Question a: We denote the observations A and B as follows:

$$(y_{1,1}, y_{1,2}) = (2, 1) \text{ and } (y_{2,1}, y_{2,2}) = (1, 2).$$

Note that there are 2 observations and therefore we can put $n = 2$. Then, the mean $\bar{y}_1$ of the first variable can be determined as follows:

$$\bar{y}_1 = \frac{y_{1,1} + y_{2,1}}{n} = \frac{3}{2}.$$

Similarly, we find that the sample mean $\bar{y}_2$ of the second component is given by

$$\bar{y}_2 = \frac{3}{2}.$$

We can determine the sample variance s_1^2 as follows:

$$s_1^2 = \frac{1}{2}\left(\left(2 - \frac{3}{2}\right)^2 + \left(1 - \frac{3}{2}\right)^2\right) = \frac{1}{4},$$

and similarly

$$s_2^2 = \frac{1}{4}.$$

Then, the standardized observations are:

$$\begin{aligned} x_{1,1} &= \frac{y_{1,1} - \bar{y}_1}{s_1} = 1, \\ x_{1,2} &= \frac{y_{1,2} - \bar{y}_2}{s_2} = -1. \\ x_{2,1} &= -1 \\ x_{2,2} &= 1. \end{aligned}$$

Question b: The covariance $s_{1,2}$ can be determined as follows

$$s_{1,2} = \frac{1}{n}\sum_{i=1}^{n}\left(x_{i,1} - \bar{x}_1\right)\left(x_{i,2} - \bar{x}_2\right).$$

Since the data is standardized, we can simplify this to

$$s_{1,2} = \frac{1}{n}\sum_{i=1}^{n} x_{i,1}x_{i,2}.$$

Using the two observations we find:

$$s_{1,2} = \frac{1 \times (-1) + (-1) \times 1}{2} = -1.$$

Then:

$$\Sigma = \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix}.$$

To determine the eigenvector $\underline{\alpha}$ and the eigenvector λ we have to solve the following equations:

$$\Sigma \begin{pmatrix} \alpha_1 \\ \alpha_2 \end{pmatrix} = \lambda \begin{pmatrix} \alpha_1 \\ \alpha_2 \end{pmatrix}.$$

This implies that:

$$\begin{aligned} \alpha_1 - \alpha_2 &= \lambda \alpha_1, \\ -\alpha_1 + \alpha_2 &= \lambda \alpha_2. \end{aligned}$$

From the first equation, we find $\alpha_2 = (1 - \lambda)\alpha_1$ and plugging this expresion in the second equation (while assuming that $\alpha_1 \neq 0$) results in:

$$\lambda(2 - \lambda) = 0.$$

We conclude that the two eigenvalues are given by

$$\lambda_1 = 2 \text{ or } \lambda_2 = 0.$$

Question c: The first principal component $\underline{\alpha}_1 = (\alpha_{1,2}, \alpha_{2,2})$ is the eigenvector which corresponds with the largest eigenvalue. From the previous question we find that the largest eigenvalue is 2, and therefore the first principal component should be determined by solving the following equations:

$$\begin{aligned} \Sigma \underline{\alpha}_1' &= 2\underline{\alpha}_1' \\ \alpha_{1,1}^2 + \alpha_{1,2}^2 &= 1, \end{aligned}$$

where the second equation follows from the normalization of the principal component. From the first equation, we find $-\alpha_{1,2} = \alpha_{1,1}$. Plugging this expression in the second equation results in:

$$\alpha_{1,1}^2 + \alpha_{1,1}^2 = 1,$$

from which we find:

$$\alpha_{1,1} = \frac{1}{\sqrt{2}}.$$

We then find that the first principal component is given by

$$\underline{\alpha}_1 = \left(\frac{1}{\sqrt{2}}, -\frac{1}{\sqrt{2}} \right).$$

The second principal component $\underline{\alpha}_2$ corresponds with the eigenvector with eigenvalue 0. Then, we find that the second principal component should be determined using

$$\begin{aligned} \Sigma \underline{\alpha}_2' &= 0 \\ \alpha_{2,1}^2 + \alpha_{2,2}^2 &= 1. \end{aligned}$$

The first equation results in $\alpha_{2,1} = \alpha_{2,2}$. Plugging this expression in the second equation gives the following principal component:

$$\underline{\alpha}_2 = \left(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}} \right).$$

Question d: The vectors are orthogonal if $\underline{\alpha}_1 \cdot \underline{\alpha}_2 = 0$. We can calculate:

$$\begin{aligned}\underline{\alpha}_1 \cdot \underline{\alpha}_2 &= -\frac{1}{\sqrt{2}} \times \frac{1}{\sqrt{2}} - \frac{1}{\sqrt{2}} \times \frac{1}{\sqrt{2}} \\ &= 0.\end{aligned}$$

Question e: There are only 2 observations and therefore we can always draw a first principal component that runs parallel with the line connecting the two observations. Therefore, the variation between the two observations is completely explained by this first component. Since there are no movements in any other direction, the second component has variance zero. The variance of the second component is given by the second eigenvalue, which is therefore equal to 0.

The total variance of the data is given by the sum of the variances:

$$\text{Total Variance} = 1 + 1 = 2,$$

where we used that the data is standardized and therefore the two variables have variance 1. We then see that the variance explained by the first component is 100%.

Exam question

1. Consider a data set with p variables, denoted by $X_1, X_2, \ldots, X_p$. We have that each variable has mean 0 and the variance of X_i is σ^2. The first principal component Z_1 is given by

$$Z_1 = \sum_{i=1}^{n} \alpha_{1,i} X_i.$$

The vector $\underline{\alpha}_1$ is defined as follows:

$$\Sigma \underline{\alpha}_1 = \lambda_1 \underline{\alpha}_1,$$

for some $\lambda_1 \in \mathbb{R}$.

Show that

$$\text{Corr}\left[Z_1, X_j\right] = \frac{\sqrt{\lambda_1}\alpha_{1,j}}{\sigma},$$

where λ_1 is the variance of Z_1.

Solution: Recall that for random variables X and Y, the correlation is defined as

$$\text{Corr}[X, Y] = \frac{\text{Cov}[X, Y]}{\sqrt{\text{Var}[X]\,\text{Var}[Y]}}.$$

Since we are interested in the calculating $\text{Corr}[Z_1, X_j]$, we will compute $\text{Var}[Z_1]$, $\text{Var}[X_j]$ and $\text{Cov}[Z_1, X_j]$.

Recall that λ_1 also has the property of being equal to $\text{Var}[Z_1]$. To see this, we write

$$\begin{aligned}
\text{Var}[Z_1] &= \text{Var}\left[\sum_{i=1}^{n} \alpha_{1,i} X_i\right] \\
&= \text{Cov}\left[\sum_{i=1}^{n} \alpha_{1,i} X_i, \sum_{k=1}^{n} \alpha_{1,k} X_k\right] \\
&= \sum_{i=1}^{n}\sum_{k=1}^{n} \alpha_{1,i}\alpha_{1,k} \,\text{Cov}[X_i, X_k] \\
&= \underline{\alpha_1}^\top \Sigma \underline{\alpha_1} && (\Sigma \text{ is covariance matrix}) \\
&= \underline{\alpha_1}^\top \lambda \underline{\alpha_1} && (\Sigma \underline{\alpha_1} = \lambda_1 \underline{\alpha_1}) \\
&= \lambda_1 && (\text{since } \underline{\alpha_1}^\top \underline{\alpha_1} = 1).
\end{aligned}$$

Furthermore, it is given that $\text{Var}[X_j] = \sigma^2$.

Next, we compute $\text{Cov}[Z_1, X_j]$:

$$\begin{aligned}
\text{Cov}[Z_1, X_j] &= \text{Cov}\left[\sum_{i=1}^{n} \alpha_{1,i} X_i, X_j\right] \\
&= \sum_{i=1}^{n} \alpha_{1,i} \,\text{Cov}[X_i, X_j] \\
&= \sum_{i=1}^{n} \alpha_{1,i} \Sigma_{ij} \\
&= \sum_{i=1}^{n} \alpha_{1,i} \Sigma_{ji}.
\end{aligned}$$

Now note that $\Sigma \underline{\alpha_1} = \lambda_1 \underline{\alpha_1}$ written out in components gives us

$$\begin{bmatrix} \Sigma_{11} & \Sigma_{12} & \cdots & \Sigma_{1n} \\ \Sigma_{21} & \Sigma_{22} & \cdots & \Sigma_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ \Sigma_{j1} & \Sigma_{j2} & \cdots & \Sigma_{jn} \\ \vdots & \vdots & \ddots & \vdots \\ \Sigma_{n1} & \Sigma_{n2} & \cdots & \Sigma_{nn} \end{bmatrix} \begin{bmatrix} \alpha_{1,1} \\ \alpha_{1,2} \\ \vdots \\ \alpha_{1,n} \end{bmatrix} = \begin{bmatrix} \lambda_1 \alpha_{1,1} \\ \lambda_1 \alpha_{1,2} \\ \vdots \\ \lambda_1 \alpha_{1,j} \\ \vdots \\ \lambda_1 \alpha_{1,n} \end{bmatrix}.$$

This implies that

$$\begin{bmatrix} \Sigma_{j1} & \Sigma_{j2} & \cdots & \Sigma_{jn} \end{bmatrix} \begin{bmatrix} \alpha_{1,1} \\ \alpha_{1,2} \\ \vdots \\ \alpha_{1,n} \end{bmatrix} = \lambda_1 \alpha_{1,j}.$$

Or in other words

$$\operatorname{Cov}[Z_1, X_j] = \sum_{i=1}^{n} \alpha_{1,i}\Sigma_{ji} = \lambda_1 \alpha_{1,j}.$$

Using the above results, we can now compute the correlation:

$$\begin{aligned}
\operatorname{Corr}[Z_1, X_j] &= \frac{\operatorname{Cov}[Z_1, X_j]}{\sqrt{\operatorname{Var}[Z_1]\operatorname{Var}[X_j]}} \\
&= \frac{\lambda_1 \alpha_{1,j}}{\sqrt{\lambda_1 \sigma^2}} \\
&= \frac{\sqrt{\lambda_1}\alpha_{1,j}}{\sigma},
\end{aligned}$$

as desired.